Bisect the location of the front and back rivet hole on the top firebox-ring rivet line; the direction of the bottom corner edges of the plate are drawn from these holes to points as dimensioned above. This completes the laying out of the plate outline.

Draw lines for the rivet lap at the front and back edge of the plate. Space-off and soap-stone circle the number of flange connecting rivets as shown on these lines. Omit the bottom rivet holes. The second hole from the bottom and the hole on the top center are punched for tack holes front and back.

The bottom hole is omitted until the flange sheets are assembled for riveting, when it is then drilled with a portable machine drill. These bottom holes are not drilled in the flat plate because of the liability of the holes drawing out of shape when the ends are scarfed.

Layout points that do not come within the plate edge are reached by clamping a strip of plate to the sheet as shown on the pattern plate layout of the crown and sides sheet, Fig. 22.

Radial Stay and Staybolt Layout

Locate the bottom staybolt line 6⅞ inches from the bottom of the plate. Measure 3 9/16 inches back of the first firebox-ring rivet hole and project this point onto the staybolt line, for the front bottom row staybolt location. Measure 78⅞ inches from the front staybolt hole to the back staybolt hole, on this line.

Scribe a short 11 15/16-inch arc, centered at the back staybolt, for locating the back developing line, as shown. Also, measure 18⅛ inches from the back set line onto the boiler center line. Draw the developing line straight from the center line, tangent to the 11 15/16-inch arc at the bottom. The developing line above the boiler center line should have the same curve as the front edge of the sheet.

Layout the front and back radial stay and staybolt lines from the edge of the plate as dimensioned.

Measure 2⅛ inches from the top center to the first radial stay line, then 28 inches from this line to the last or outside radial stay line. Measure 44⅞ inches, on the developing line, from the last radial stay line. Square this point 90 degrees to the developing line. This line should intersect the center of the back bottom staybolt hole, which has been laid out on the bottom staybolt line. Space off the 7 equal radial stay and 11 equal staybolt spaces on the back developing line. Project a point for each one of the staybolt divisions, 90 degrees to the developing line, onto the back staybolt line.

The radial stay and staybolt dimensions at the front end are measured along the front edge of the plate, which is also the front developing line. The developing line is, of course, extended below the tube sheet break in the sheet to the bottom staybolt and as dimensioned at the base line. The front staybolt divisions are also projected at right angles to the developing line onto the front staybolt line.

The longitudinal radial stay lines are always drawn directly from the divisions as spaced on the developing lines.

The longitudinal staybolt lines are, of course, drawn from the points that have been projected on the front and back staybolt lines.

The distance between the front and back radial stays, on each longitudinal line, is gaged with a metal strip. The number of equal stay spaces, as shown on the plate development drawing, are spaced off on the metal strip (four lines on each strip) and then marked onto the plate.

The main staybolt spaces are also spaced on a metal strip and then marked on the plate. The broken up staybolt spaces at the back and front corners of the plate are spaced off directly on the plate. Also, the transverse broken-up staybolt spaces in the bottom part of the plate are spaced off on the plate.

The radial stay and staybolt divisions as spaced on the metal strips are reserved for the roof sheet layout.

Soap-stone circle the boundary and broken up spaces of stay centers as shown.

This completes the laying out of the sheet. The work should now be checked thoroughly: then center punch each rivet and stay hole center; block off and mark; white lead the rivet and stay drilling sizes, after which the plate is transferred to a punching machine for punching the tack holes and for punching out the back end and bottom surplus stock as shown. Punching the surplus plate into smaller pieces, facilitates trimming the plate after it is drilled.

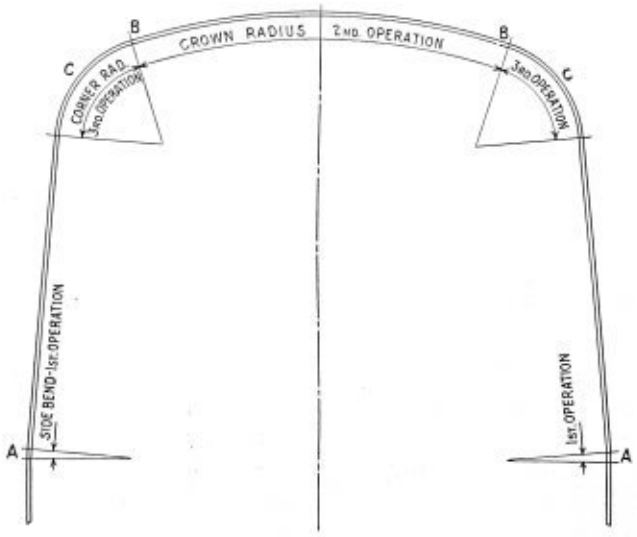


Fig. 23.—Cross section of a one-piece crown and sides sheet rolled into shape, showing locations of the rolling operations

The plate is now returned to the laying out floor for center punching the tack hole locations in the duplicate plates; also, for marker punching lines to indicate where the surplus pieces of plate are to be punched out. The length of these surplus pieces is important when the same are to be used for boiler reinforcing liners, and should be laid out on the pattern plate by the layer out to include the size of liner plate required.

Machine Operations

The plates are now removed to the multiple drilling machines where five plates, including the pattern plate, are bolted through the tack holes and further clamped together along the edges with heavy screw clamps.

When the drilling of the plates has been completed a rough line of the rivet lap plus ½ inch, is marked all round the edge of the plate. The front and back edges of the plate are then removed to this line with a cutting-off punch. The bottom or firebox-ring edges are sheared off with the straight cutting shearing machine. Sharp corners are punched out nearer the finish line with a round punch.

The edges of this sheet are required to be beveled for calking. A soap-stone line of the rivet lap or the actual finished outline of the plate is now marked all around the duplicate plates similar to the outline that has been center punched around the pattern plate.